Class Marks Analyzer

Test Cases Report | PS No. 11 | Python 3 + NumPy

Overview

This document lists the test cases used to check the Class Marks Analyzer (`class_marks_analyzer.py`). Marks are entered as numbers separated by spaces and must be from 0 to 100. The pass mark is 40 and standard deviation is the population value (`np.std` default). Every output below was produced by running the program.

Summary

#	Scenario	Input	Key result	Status
1	Normal Input	40 50 60 70 80	Mean 60.00, Median 60.00, StdDev 14.14, Passed 5 of 5	**PASS**
2	Boundary / Minimum Input (one student)	75	Mean 75.00, Median 75.00, StdDev 0.00	**PASS**
3a	Invalid Input: non-numeric value	50 abc 70	Error: only numbers allowed	**PASS**
3b	Invalid Input: marks out of range	50 10 570	Error: marks must be 0 to 100	**PASS**
4	Different Valid Scenario	25 50 75 100	Mean 62.50, StdDev 27.95, Passed 3, Failed 1, 75.00%, Above avg 2	**PASS**
5	Extra: Empty Input	(empty)	Error: enter at least one mark	**PASS**

Detailed Test Cases

Test Case 1: Normal Input

INPUT

```
Marks: 40 50 60 70 80
```

OUTPUT

```
Mean: 60.00 | Median: 60.00
Max: 80 | Min: 40 | StdDev: 14.14
Passed: 5 of 5 | Failed: 0 | Pass percentage: 100.00%
Above average: 2 students
```

Test Case 2: Boundary / Minimum Input (one student)

INPUT

```
Marks: 75
```

OUTPUT

```
Mean: 75.00 | Median: 75.00
Max: 75 | Min: 75 | StdDev: 0.00
Passed: 1 of 1 | Failed: 0 | Pass percentage: 100.00%
Above average: 0 students
```

Test Case 3a: Invalid Input: non-numeric value

INPUT

```
Marks: 50 abc 70
```

OUTPUT

```
Invalid marks 'abc'. Enter numbers only.
```

Test Case 3b: Invalid Input: marks out of range

INPUT

```
Marks: 50 10 570
```

OUTPUT

```
Marks must be between 0 and 100.
```

Test Case 4: Different Valid Scenario

INPUT

```
Marks: 25 50 75 100
```

OUTPUT

```
Mean: 62.50 | Median: 62.50
Max: 100 | Min: 25 | StdDev: 27.95
Passed: 3 of 4 | Failed: 1 | Pass percentage: 75.00%
Above average: 2 students
```

Test Case 5: Extra: Empty Input

INPUT

```
Marks:
```

OUTPUT

```
Please enter at least one mark.
```

Result: all 6 test cases passed.